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A NEW CLASS OF MARKET SHARE MODELS

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Applications of market share models which (implicitly) rely on Luce's choice axiom have been widely criticized because they cannot account for the effects of differential product substitutability and product dominance. Three types of choice models—Tversky's Elimination-By-Aspects model, Tree Models, and Generalized PROBIT—have been offered as solutions to the problems identified with the Luce model, but they each suffer from limitations which have prevented their widespread application in marketing contexts. Tversky's elegant EBA model has not been widely used because it requires a large number of parameters and no special-purpose parameter estimation software has yet emerged. Tree models have been offered as more parsimonious special cases of EBA, but they are more restrictive in that they presume: (1) that products, and the process of choosing from among them, can be characterized in terms of hierarchical, attribute-based trees; and (2) that the aspects governing choice are well-known. Generalized PROBIT can paramorphically handle the problems with the Luce model, but parameter estimation software has proved problematic because it cannot guarantee a globally optimum solution. This paper proposes a new class of market share models. Rather than model the choice process explicitly, the new models simply scale the effects competing products have on each other's market share. These competitive effects are scaled in the context of a class of market-share models which: (1) do not assume a tree-like structure for the competing products; (2) do not presume any a priori knowledge about the attributes governing choice; (3) are characterized in terms of parameters that can be estimated using ordinary least-squares; and (4) provide clear managerial insight into the sources of competition. The paper begins with a brief review of previous work. Following the review, the paper offers a theorem and proof which guarantees the existence of the new class of models. The empirical validity and strategic utility of the models are demonstrated using two separate sets of data.

Introduction

Background

The development and application of models of market share has been a primary activity in the fields of marketing, management science and applied economics for the last two decades: Huff (1962); Kuehn and Day (1962); Hlavac and Little (1966); Pessemier, Burger, Teach and Tigert (1971); Nakanishi (1972); McFadden (1973); Nakanishi, Cooper and Kassarjian (1974); Shocker and Srinivasan (1974); Wind, Jolly and O'Connor (1976); Hauser and Urban (1978); Punj and Staelin (1978); Reibstein (1978); Silk and Urban (1978); Flath and Leonard (1979); Gensch and Recker (1979); Batsell and Lodish (1981). Although different in functional form, data requirements, parameter estimation, and level of application, these methodologies have in common a single crucial assumption:

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for every product there exists a scale value which reflects its utility and the market share for that product is proportional to its scale value. Because it can be demonstrated that the "single scale value assumption" is implied by Luce's axiom on choice behavior (Luce 1959), we characterize models relying on this assumption as Luce-based. Bell, Keeney and Little have captured the assumption in the context of a market share theorem (1975).

Problems with Luce-based Models of Market Share

Recognizing limitations in the applicability of his model, Luce was the first to caution against applying his axiom on choice to complex choices involving many different types of alternatives (Luce 1959, pp. 7 and 133). Rather, he argues it might be expected to hold only for simple choices. Similarly, in their review of conjoint analysis in consumer research Green and Srinivasan note that the assumption which underlies the Luce axiom "may not be a realistic assumption in many consumer behavior contexts" (1978, p. 113).

There exist three well-documented problems with the Luce model. In describing each of the problems below we will assume the perspective of an individual making repeated choices from different sets of offered products. The same problems can arise when the market shares are derived from choices made by a homogeneous segment or an entire market.

The first type of problem with the Luce model has been labeled the similarity (or substitutability) problem and was first noted by Debreu (1960). To illustrate the problem consider an individual repeatedly choosing from a two-product market composed of Coke and 7-Up where the respective products each capture 50% of his choices. Now suppose a new product called Pepsi is introduced into his choice set and he likes it just as much as he does either Coke or 7-Up. The Luce model would "predict" equal market shares ($33\frac{1}{3}\%$) for all three beverages. If, however, he perceives Pepsi and Coke to be very substitutable for each other and neither very substitutable for 7-Up, then a more reasonable expectation would be that Pepsi would pull disproportionately more from Coke's share of choices than from those of 7-Up. If true, 7-Up's share in the three-product market would be larger than either Coke or Pepsi and might even remain at 50%. Empirical evidence supporting this phenomenon has been provided by Becker, DeGroot and Marschak (1963), Rummelhart and Greeno (1971), Bass, Pessemier and Lehmann (1972), and Huber and Puto (1983).

The second problem with the Luce model has been characterized as the dominance problem. To illustrate, consider a different example using the same 3 products. Suppose that on the occasions when only Pepsi and 7-Up are available their shares are 0.50, 0.50, respectively, and that when only Coke and 7-Up are available their shares are 0.50, 0.50, respectively. Also suppose, however, that Pepsi "dominates" Coke. Thus, when choosing from sets containing only a cola or a 7-Up the shares are 0.50, 0.50 as described above, but when choosing from the set containing only Pepsi and Coke the shares are 0.95 and 0.05, respectively, because Pepsi is almost always chosen over Coke. Such data, although reasonable, could not be modeled using the Luce axiom. Empirical evidence supporting this phenomenon has been provided by Rummelhart and Greeno (1971) and Huber and Sewall (1978).

The third problem with the Luce model occurs when choice probabilities violate regularity. To illustrate, consider an individual choosing from two different shirts on a shelf where he likes the two shirts equally and their respective choice probabilities are thus 0.5, 0.5. Now suppose a clerk deposits a third shirt on the shelf which is identical to one of the original two. If our individual wanted to minimize the chances of wearing a shirt identical to someone else's the probability that he would choose the unique shirt from the set of three could be expected to be well over 0.5. This result violates regularity because regularity requires that the introduction of a new alternative to a set will never *increase* the probability of choice for an alternative already in the set. Empirical evidence

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demonstrating violations of regularity have been provided by Huber, Payne and Puto (1982).

Previous Attempts to Solve the Problems

Awareness of the substitutability and dominance problems has been widespread and attempts to solve them have resulted in three types of approaches. The first approach is Tversky's Elimination-By-Aspects (EBA) model of choice behavior (Tversky 1972). Assuming each alternative in some consideration set can be viewed as a collection of aspects, Tversky proposes that choice is a covert elimination process. At each stage in the choice process, an aspect is selected with probability proportional to its measure, and alternatives not characterized by the aspect are eliminated. The process continues until only one alternative remains. It is also possible to describe the EBA model without reference to the aspects characterizing the products. In this description one simply has a scale value corresponding to the unique aspects of each product, a scale value corresponding to those aspects shared by each pair of products, each triple of products, . . . etc. Tversky's EBA model provides an elegant solution to the substitutability and dominance problems, is well-grounded in theory, and can be clearly described in terms of an individual's actual choice process. The EBA model has not enjoyed widespread application in Marketing, however, because it requires a large number of parameters (for N alternatives, EBA requires $2^N - 3$ parameters) and no special-purpose parameter estimation software has yet emerged.[1]

The second category of approaches designed to handle problems with the Luce model have involved the use of tree-like structures (Tversky and Sattath 1979; McFadden 1981; Urban, Johnson and Hauser 1984). In these approaches each alternative is represented as a collection of aspects, the aspects are assumed to have a tree structure, and choice is described as a covert hierarchical process of elimination. The parameter estimation problem has been solved by using maximum likelihood methods, but these approaches require specification of the aspects and tree structures a priori. These approaches should be quite useful in applications where the alternatives can be portrayed in a tree-like structure, choice can be reasonably characterized as hierarchical in nature, and the aspects governing choice are well known. In markets where the product offering is not conformable to a tree-like structure or the aspects governing choice have not already been clearly defined, application of the tree models can prove troublesome. Even when tree structures seem appropriate, some problems arise in discriminating between trees. It can happen, for example, that two different tree structures can provide indistinguishable fits to the data and yet produce distinguishable estimates for market shares for new products. Neither the EBA model nor the tree-like models can handle violations of regularity in choice data.

The third class of approaches falls under the category of multinomial probit. An excellent review and generalization of these approaches has been provided by Currim (1982). Essentially, this approach relaxes constraints on the error-structure which then permits a greater dependence between the effects of alternatives than is possible under the Luce model. In this way Generalized PROBIT can handle problems with substitutability, dominance, and violations of regularity. Approaches falling under this third category use parameter estimation software which rely on maximum likelihood search routines that can be quite time-consuming and do not guarantee global optimum (Currim 1982).

[1] Although Tversky (1972) used STEP-IT (a general-purpose maximum-likelihood routine) to estimate EBA parameters, no one has developed and disseminated software specifically designed for EBA parameter estimation. The availability of efficient, reliable, and easy-to-use parameter estimation software is crucial for the wide-spread acceptance of new models in applied settings. We believe that the absence of such software for EBA has been a contributing factor to its lack of use.

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A New Approach

This paper proposes a new class of market share models which solve the substitutability problem, the dominance problem, and problems associated with violations of regularity. Rather than model the choice process explicitly, the new models simply scale the effects competing products have on each other's market share. These competitive effects are scaled in the context of a class of market-share models which:

(i) do not require that competing products be conformable to a tree-like structure;
(ii) do not necessarily assume any a priori knowledge about the aspects governing choice;
(iii) are characterized in terms of parameters that can be estimated using ordinary least-squares; and
(iv) provide clear managerial insight into the sources of competition.

An Intuitive Description

In this section we offer a brief intuitive description of the model proposed in this paper. Assume we have obtained market shares for all 11 possible subsets of 4 products (six 2-product sets, four 3-product sets, and the one set containing all 4 products). First, consider the 2-product set composed of products i and j. The ratio of i's share to j's share provides an important measure of how one product does versus the other in a simple one-on-one context when the presence of other products cannot confound the results. (Call this ratio a second-order effect because it involves 2 products, i and j.) Now suppose we consider the ratio of i's share to j's share in the 3-product set composed of i, j, and k. If, in the addition of k, k pulls from i and j in roughly the same proportion, the ratio of i's share to j's share in the 3-product set could be approximated by the ratio observed in the 2-product set. However, if k, for whatever reason, pulls proportionally more from i's share than j's share, the ratio of i's share to j's share in the 3-product set will *not* be the same as in the 2-product set—it will be smaller. We could, however, represent the ratio of i's share to j's share in this 3-product set as the *product of the original second-order effect and a measure of the degree to which k pulled from i versus j.* (Call this new measure a third-order effect.) In a sense, we have decomposed the ratio of i's share to j's share in the 3-product set into a component due to the competition between i and j (a second-order effect) and a component due to the differential way in which k draws from i versus j (a third-order effect).

But now suppose we consider the ratio of i's share to j's share in the 4-product set composed of i, j, k and l. It may be possible to approximate this ratio by the product of three components—the second-order effect of i versus j, the third-order effect of k on i versus j, and the third-order effect of l on i versus j. If our approximation of this new ratio does not come sufficiently close, then we must add a new component representing the joint effect of k and l operating on i versus j (a fourth-order effect) in addition to the 3 simpler effects described above.

The basis for the class of models proposed in this paper is the expectation that, in general, the lower-order effects will be sufficient to approximate the ratio of market shares for any 2 products in any set.[2] In the simplest case, second-order effects may be sufficient. (In this case the Luce choice axiom is satisfied.) In the next simplest case second-order and third-order effects will be sufficient and higher-order effects will not be necessary. Failing this, fourth-order effects may need to be introduced, . . . etc. In the next section we introduce the idea more formally.

[2] Some reviewers have observed that there exists an analogy between our class of market-share models and analysis-of-variance with interaction effects.

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Definition of Terms

Let $T = \{1, 2, \ldots, N\}$ be a set of alternative products. For each subset $A \subset T$ let $P_A(i)$ denote the probability that alternative i will be chosen when the set A is the set of available alternatives.

In what follows we will assume no a priori relationship between the probabilities P_A. We will only assume that $P_A(i) \neq 0$ for all $i \in A$ and for all $A \subset T$.

For each nonempty subset $A \subset T$ and for $i, j \in A$ we define

$$\beta_{ij}^A = \ln\left(\frac{P_A(i)}{P_A(j)}\right).$$

Clearly,

$$\beta_{ii}^A = 0 \qquad \text{for all } i, \tag{1}$$

$$\beta_{ij}^A + \beta_{ji}^A = 0 \qquad \text{for all } i, j, \tag{2}$$

$$\beta_{ij}^A + \beta_{jk}^A + \beta_{ki}^A = 0 \qquad \text{for all } i, j, k. \tag{3}$$

THEOREM.

There exist unique numbers α_{ij}^I defined for $i, j \in T$, $I \subset T$ with $I \cap \{i, j\} = \varnothing$ such that for every subset $A \subset T$, (4)

$$\beta_{ij}^A = \sum_{I \subset [A - \{i,j\}]} \alpha_{ij}^I. \tag{5}$$

In fact α_{ij}^I is given by

$$\alpha_{ij}^I = (-1)^{\#(I)} \sum_{J \subset I} (-1)^{\#(J)} \beta_{ij}^{ijJ} \tag{6}$$

where $\#(I)$ denotes the number of elements in the set I.

To facilitate readability, we offer the proof in Appendix A.

Interpretation

First, if $A = \{i, j\}$, $I = \varnothing$, and we have $\beta_{ij}^A = \beta_{ij}^{ij} = \alpha_{ij}^{\varnothing}$. In the case where $A = \{i, j, k\}$ (5) becomes $\beta_{ij}^{ijk} = \alpha_{ij}^{\varnothing} + \alpha_{ij}^k$. Thus, $\alpha_{ij}^{\varnothing}$ is the natural logarithm of the ratio of i's share to j's share, and α_{ij}^k measures the effect of the presence of alternative k on the natural logarithm of that ratio. In other words, α_{ij}^k is the measure of the effect of the presence of alternative k on the competition between alternatives i and j.

If $A = \{i, j, k, l\}$ (5) becomes

$$\beta_{ij}^{ijkl} = \alpha_{ij}^{\varnothing} + \alpha_{ij}^k + \alpha_{ij}^l + \alpha_{ij}^{kl}.$$

Thus, α_{ij}^{kl} is the measure of the effect of the pair kl on the competition between i and j with the effect of the individual products k and l already accounted for by α_{ij}^k and α_{ij}^l. In effect, α_{ij}^{kl} measures an interaction of the quadruple $\{i, j, k, l\}$. We call α_{ij}^{kl} a *fourth order effect.*

In general α_{ij}^I measures the effect of the products in the set I on the competition between i and j with the simpler effects of all proper subsets $J \subsetneq I$ already accounted for in (5) *by the factors α_{ij}^J. In general, if $n = \#(I \cup \{i, j\})$ we will call α_{ij}^I an nth order effect.*

Additional Notation

For convenience we will set

$$\alpha_{ij}^I = 0, \qquad \text{if} \qquad I \cap \{i, j\} \neq \varnothing, \qquad \text{and}$$

$$\alpha_{ii}^I = 0 \qquad \text{for all } I, i.$$

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With this convention, (5) becomes

$$\beta_{ij}^{A} = \sum_{I \subset A} \alpha_{ij}^{I}. \tag{8}$$

Relationship Between Parameters

The factors α_{ij}^{I} are not independent. The relations among them are implied by (1), (2), and (3) and are specified in the next result.

PROPOSITION 9.

$$\alpha_{ij}^{I} + \alpha_{ji}^{I} = 0. \tag{10}$$

If $I \cap \{i, j, k\} = \varnothing$, *then*

$$\alpha_{ij}^{I} + \alpha_{jk}^{I} + \alpha_{ki}^{I} + \alpha_{ij}^{kI} + \alpha_{jk}^{iI} + \alpha_{ki}^{jI} = 0. \tag{11}$$

Again to facilitate readability, the proof is offered in Appendix B.

Number of Free Parameters

For each $\varnothing \neq A \subset T$ there are $\#(A) - 1$ free parameters in the specification of P_A. Hence there are

$$\sum_{k=2}^{N} \binom{N}{k}(k-1) = (N-2)2^{N-1} + 1$$

free parameters in the specification of $\{P_A \mid \varnothing \neq A \subset T\}$.

Potential for Application

The essence of the proposed class of market share models is contained in (8). The theorem (4) demonstrates that the logarithm of the ratio of two market shares (i to j) for any set (A) of competing products can be decomposed into the sum of a set of effects, (α_{ij}^{I}) each of which is of an increasingly higher order.

The effects represented by the α's each have a clear behavioral interpretation of managerial significance. For example, when product k is added to the set $\{i, j\}$:

(1) if $\alpha_{ij}^{k} < 0$, k pulls from i's share proportionally more than it pulls from j's share;
(2) if $\alpha_{ij}^{k} = 0$, k pulls from both i's share and j's share in the same proportion;
(3) if $\alpha_{ij}^{k} > 0$, k pulls from j's share proportionally more than it pulls from i's share; and,
(4) if $\alpha_{ij}^{k} = -\alpha_{ij}^{l}$, then k pulls from i versus j in the same proportion as l pulls from j versus i.

The number of free parameters needed to describe the probabilities $P_A(i)$ is rather large (for N products there are $(N-2)2^{N-1} + 1$ parameters). When the parameters chosen to describe the $P_A(i)$'s are the α_{ij}^{I}'s, the parameters have the specific interpretation indicated earlier. It is quite tempting, however, to hypothesize that the lower order alphas will explain most of the competitive effects, and that the higher order alphas will be less important. Accordingly, we define the *nth order model* as

$$\beta_{ij}^{A} = \ln P_A(i) - \ln P_A(j) = \sum_{I \subset A} \alpha_{ij}^{I}$$

with $\alpha_{ij}^{I} = 0$ if $\#(I \cup \{i, j\}) > n$. Of course we also require that (10) and (11) are satisfied. With this definition, the 2nd order model is just the Luce model. The models of orders ≥ 3 are new.

Suppose, for example, that we have market shares for all possible subsets of 4 products. The $P_A(i)$'s are then completely described by the 4th order model, which has 17 free parameters. It may be possible to approximate the probabilities, however, by the 3rd

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order model with 11 free parameters, or by the 2nd order model (the Luce model) with 3 independent parameters. Thus, in an applied setting, with a large number of products, the general-case solution provided by (8) may be undesirable because of the large number of parameters. (In fact, the general-case solution described by (8) is a "saturated model" in that there are precisely as many independent parameters as there are independent observations.) But it may be possible to approximate (8) by lower order models which involve many fewer parameters. Furthermore, the hierarchical nature of these models permits statistical tests of the relative significance of the models of different orders. In this manner the researcher can make explicit tradeoffs between the agreement with the data and the level of parsimony of the models. The point here is that it might be possible to explain most of the variance in the redistribution of market shares across subsets with a relatively parsimonious lower order model.

Parameter Estimation

We have found it convenient to estimate the α_{ij}^I's indirectly through an equivalent set of parameters $\{s_i^I\}$. In Appendix C we demonstrate that (8) is equivalent to:

$$P_A(i) = \frac{\prod_{I \subset A} e^{s_i^I}}{\sum_{k \in A} \prod_{I \subset A} e^{s_k^I}}, \qquad \text{where} \tag{12}$$

$$s_i^I = 0 \qquad \text{for } i \notin I; \tag{13}$$

$$\sum_{i \in I} s_i^{iI} = u_I; \qquad \text{and} \tag{14}$$

u_I is any preassigned number. Note that if $s_i^I = 0$ for all $I \neq \varnothing$, and all i, (12) reduces to the Luce model.

Although (12) is nonlinear in its parameters, in Appendix D we demonstrate that the simple log transformation first used by Nakanishi and Cooper (1974) yields

$$\ln\left(\frac{P_A(i)}{\bar{P}_A}\right) = \left(\frac{N_A - 1}{N_A}\right)\left(\sum_{I \subset A} s_i^I\right) + \left(-\frac{1}{N_A}\right)\left(\sum_{\substack{I \subset A \\ k \neq i}} s_k^I\right) \qquad \text{where} \tag{15}$$

$\bar{P}_A$ = the geometric mean of the probabilities in A; and
N_A = the number of alternatives in A.

Because (15) is linear in its parameters, least squares can be used to estimate the s_i^I's. Once the estimates for the s_i^I's are obtained, the α_{ij}^I's are given by

$$\alpha_{ij}^I = s_i^{iI} + s_i^{ijI} - s_j^{jI} - s_j^{ijI} \qquad \text{if} \qquad I \cap \{i, j\} = \varnothing. \tag{16}$$

Finally, in Appendix E we demonstrate parameter estimation on a hypothetical set of data. For more details on least squares parameter estimation for this class of models see Nakanishi and Cooper (1974) and Batsell and Krieger (1979). (We note that there may be maximum likelihood approaches to estimating the parameters for (12), and therefore (8), as well.)

Brief Comparison with EBA

Perhaps the choice model most similar to the models proposed in this paper is Tversky's EBA model which was briefly described in the introduction. In this section we offer a very brief comparison between the EBA model and the class of models proposed here.

As noted in the introduction, EBA can be described as both a model of the actual choice process *and* a scaling model. The models we propose here are only scaling models in the sense that they do not assume some underlying model of the choice process itself.

Nevertheless, the two approaches are similar in that they are both characterizable by

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EXHIBIT 1

Differences in the Number of Independent Parameters for Tversky's EBA Model and the Model Proposed in this Paper Assuming 5 Products

Level in Hierarchy	EBA Model	Number of Independent Parameters	Our Model	Number of Independent Parameters
First Level	scale values on objects	5 − 1 = 4	second-order alphas	4
Second Level	scale values on objects, and object pairs	15 − 1 = 14	second- and third-order alphas	19
Third Level	scale values on objects, object pairs, and object triples	25 − 1 = 24	second-, third-, and fourth-order alphas	39
Fourth Level	scale values on objects, object pairs, object triples and sets of four	30 − 1 = 29	second-, third-, fourth- and fifth-order alphas	49

natural hierarchies. The scale value description of EBA (where reference to underlying aspects is not necessary) possesses a natural hierarchy—scale values for objects, object pairs, object triples, . . . etc. The models proposed here also have second order effects—a scale value for each pair of objects—, third order effect—a scale value for each object k crossed by each pair of objects $\{i, j\}$ where $k \neq i, j$—, . . . , etc.

Despite the similarity of hierarchies, the two approaches are different in two important ways. First, although the EBA model is additive, the models proposed here are multiplicative—they are additive in logs only. Second, as is made evident by Exhibit[1], the number of independent parameters at each level in the two hierarchies is quite different.

Application and Empirical Results

This section describes the application of the proposed models to data from two separate experiments. In the first study the models are applied to individual-level data, and in the second, to group-level data. The purpose of the section is two-fold. First, it is intended as a demonstration of the empirical validity of the proposed models. Second, and perhaps more important, this section demonstrates several insights into the sources of competition that are suggested by the parameters in the models.

Individual-Level Application

The Experiment and the Data. In this study the models are applied to data resulting from the repeated choices of one individual. The subject, a student volunteer, had a regular habit of taking a study break each weeknight. During the break he would proceed to a row of vending machines in the basement of his dorm and purchase a snack—usually one of the following five products—Yodels, Twinkies, Devil Dogs, Milky Way, and Three Musketeers. Over a period of twelve evenings, he was asked to modify his behavior in the following way. On each evening he was provided with a list of *all* 26 *possible subsets of the* 5 *products*—all subsets containing 2 or more of the 5 products. Before proceeding to the vending area, he indicated, for each of the 26 subsets, the snack in that subset he would most like to buy that evening. In making his choice from each

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subset he was asked to imagine that the vending machines were out of all snacks but those listed in the set.

Finally, he was provided with 12 random numbers (1 for each evening) between 1 and 26. After making his choices from each of the 26 subsets, the random number for the evening was used to designate a single subset from the 26. To enhance the importance of the a priori choice from each subset, the subject agreed to actually purchase the snack he had chosen from the subset indicated by the random number for the evening.

Expected Results

Note that the first three products are small cake-like snacks and the last two are candy bars. Thus, we would expect the products to be differentially substitutable, with Yodels, Twinkies, and Devil Dogs pulling proportionally more from each other's market shares than they would from Milky Way and Three Musketeers; similarly, Milky Way and Three Musketeers should pull proportionally more from each other's market shares than either would from the first three products.

Furthermore, if the products are differentially substitutable there should be a significantly better fit for the 3rd order model compared to the second order model. Finally, it would be a particularly nice result if the fourth order model did not fit significantly better than the 3rd order model because this would suggest the possibility that a relatively parsimonious model might be sufficient to capture the majority of the differential interproduct competitive effects.

Actual Results

Table 1 contains the results of fitting and testing alternative forms of the proposed models to the individual-level data. First note that the 2nd order model (equivalent to the Luce model) fit significantly better ($p < 0.01$; adjusted $R^2 = 0.2683$) than a naive model of equal shares in all choice sets. But also note that the introduction of the 3rd order α's provided a significant better fit ($p < 0.0001$; adjusted $R^2 = 0.8611$) to the data than did the 2nd order model. When comparing the 3rd order model to the 4th order model, however, the difference in fit is significant only at the $p = 0.0971$ level. Choosing between the third and fourth order models would depend on the researcher's goals, but one could make a rather strong case for adopting the 3rd order model and proceeding to investigate the values of the alphas.

Insights into Competitive Structure from the Third-Order Solution

Table 2 contains the parameter estimates for the fourth, third, and second-order models, respectively. In this section we first describe some of the insights into competitive structure that can be obtained by examining the *3rd order solution.* Then we discuss the effects of

TABLE 1

Comparison of Fit and Statistical Tests of Alternative Models: Individual-Level, Snack Data

Order of Model	Number of Independent Parameters	Number of Independent Observations	Error Sum of Squares	R-Square	Adjusted R-Square	F with Previous Level	Probability for F
2	4	49	4.470	0.3141	0.2683	6.88	$p < 0.01$
3	19	49	0.566	0.9132	0.8611	13.81	$p < 0.0001$
4	39	49	0.104	0.9841	0.9235	2.22	$p < 0.0971$
5	49	49	0.0	1.0	1.0	NA*	NA*

* With the 5th order solution the number of independent parameters equals the number of independent observations, the error sum-of-squares is 0, and the F statistic cannot be computed.

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TABLE 2

Parameter Estimates from Fourth-Order, Third-Order, and Second-Order Models: Individual-Level Snack Data

Fourth-Order Solution—Values of α_{ij}^{I} For:

	And I Equal To:															
i/j Equal To:	{∅}	{1}	{2}	{3}	{4}	{5}	{1, 2}	{1, 3}	{1, 4}	{1, 5}	{2, 3}	{2, 4}	{2, 5}	{3, 4}	{3, 5}	{4, 5}
1/2*	0.002	—	—	−0.977	0.454	0.259	—	—	—	—	—	—	—	0.043	0.210	−0.349
1/3	−0.165	—	−0.408	—	−0.232	−0.215	—	—	—	—	—	0.331	0.297	—	—	0.021
1/4	0.767	—	−0.605	−1.033	—	0.427	—	—	—	—	0.141	—	0.149	—	0.205	—
1/5	0.361	—	−0.598	−0.569	0.396	—	—	—	—	—	0.023	0.208	—	0.203	—	—
2/3	0.088	0.313	—	—	−0.175	−0.186	—	—	−0.223	−0.202	—	—	—	—	—	0.370
2/4	0.664	−0.956	—	−0.664	—	0.659	—	0.707	—	0.007	—	—	—	—	−0.005	—
2/5	0.313	−0.811	—	−0.312	0.544	—	—	0.532	−0.044	—	—	—	—	0.160	—	—
3/4	0.308	−0.176	−0.221	—	—	0.617	−0.165	—	—	0.209	—	—	−0.148	—	—	—
3/5	0.650	−0.478	−0.551	—	0.632	—	0.087	—	0.178	—	—	−0.123	—	—	—	—
4/5	−0.280	−0.158	−0.186	0.636	—	—	0.252	−0.174	—	—	−0.118	—	—	—	—	—

Third-Order Solution—Values of α_{ij}^{I} For:

	And I Equal To:					
i/j Equal To:	{∅}	{1}	{2}	{3}	{4}	{5}
1/2	0.056	—	—	−0.654	0.156	0.080
1/3	−0.357	—	−0.105	—	−0.003	−0.039
1/4	0.691	—	−0.455	−0.997	—	0.642
1/5	0.318	—	−0.579	−0.543	0.655	—
2/3	0.243	−0.106	—	—	−0.159	−0.119
2/4	0.475	−0.451	—	−0.342	—	0.562
2/5	0.157	−0.554	—	(0.111)	0.499	—
3/4	0.399	−0.345	−0.350	—	—	0.681
3/5	0.620	−0.449	−0.474	—	0.658	—
4/5	−0.257	−0.104	−0.124	0.454	—	—

Second-Order Solution—Values of α_{ij}^{I} For:

	And I Equal To:
i/j Equal To:	{∅}
1/2	−0.166
1/3	−0.262
1/4	0.242
1/5	0.116
2/3	−0.096
2/4	0.408
2/5	0.282
3/4	0.504
3/5	0.377
4/5	−0.126

*1 = Yodels
2 = Twinkies
3 = Devil Dogs
4 = Milky Way
5 = Three Musketeers

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differential competition that lead to changes in parameter estimates for the 2nd order versus 3rd order solution.

First consider the second order parameter, $\alpha_{ij}^{\varnothing}$. In the general case model, $\alpha_{12}^{\varnothing}$ would simply be the log of the ratio of product 1's share to that of product 2 in the 2-alternative set containing only products 1 and 2. If, however, the 3rd order model is adopted because the higher order models do not provide significantly better fits, then $\alpha_{12}^{\varnothing}$ represents the best estimate (in a least-squares sense) of the log of the ratio of i's share to j's share across all subsets given that all third-order competitive effects are also captured in the model. Note that the $\alpha_{ij}^{\varnothing}$'s will be a different and better indication of the preference for product i versus product j than simply taking the overall average of the logs of the ratios of their shares across all sets because the $\alpha_{ij}^{\varnothing}$ is estimated with the significant 3rd order effects present in the model. The presence of all significant higher-order parameters prevent any differential competitive effects from distorting the second-order alpha as an index of preference for i versus j.

If we take $\alpha_{ij}^{\varnothing} > 0$ to mean that i is preferred to j, then Table 2 indicates that the subject was close to indifferent between Yodels and Twinkies ($\alpha_{12}^{\varnothing} = 0.056$), that he preferred Devil Dogs to Yodels ($\alpha_{13}^{\varnothing} = -0.357$), but rather strongly preferred Yodels to Milky Way ($\alpha_{14}^{\varnothing} = 0.691$). Furthermore, the 2nd order alphas indicate that he strictly prefers the 3 cake-like products to the 2 candy bars.

Now consider the 3rd order alphas (α_{ij}^{k}) for $k = 1$. Note that Yodels pulled from Twinkies market share more than it did from that of either Milky Way ($\alpha_{24}^{1} = -0.451$) or Three Musketeers ($\alpha_{25}^{1} = -0.554$). Also as expected, Yodels pulled from Devil Dogs more than it did from either Milky Way ($\alpha_{34}^{1} = -0.345$) or Three Musketeers ($\alpha_{35}^{1} = -0.449$).

Overall, there are 18 3rd order alphas (α_{ij}^{k}) for which we have clear a priori expectations. Each of these alphas have been underlined in the third-order solution in Table 2. Whenever the product k is from the same product group as j, α_{ij}^{k} should be greater than 0. Of the 18 alphas, only 1 ($\alpha_{25}^{3} = 0.111$) is different than expected. It suggests that when the product Devil Dogs is introduced into the set with Twinkies and Three Musketeers, it takes proportionately more share from Three Musketeers than Twinkies. The other 17 alphas for which there exist a priori expectations are consistent with those expectations.

Changes in Parameter Estimates for Different Order Solutions

We now briefly consider some of the changes observed in the 2nd order alphas when the 2nd order solution is compared to the 3rd order solution. Notice that $\alpha_{12}^{\varnothing}$ goes from -0.166 to 0.056 and $\alpha_{23}^{\varnothing}$ goes from -0.096 to 0.243 when comparing the 2nd order solution with the 3rd order solution. The 3rd order alphas from the 3rd order solution help resolve the differences. In the first instance $\alpha_{12}^{\varnothing} = -0.166$ suggests the subject preferred Twinkies to Yodels but $\alpha_{12}^{\varnothing} = 0.056$ suggests the subject either prefers Yodels to Twinkies or that he is indifferent between them. The key is $\alpha_{12}^{3} = -0.654$ which indicates that Devil Dogs pull substantially more from Yodels than Twinkies. Thus, in sets where Yodels, Twinkies, and Devil Dogs are all present, Devil Dogs pull so much more from Yodels than Twinkies that a model without the 3rd order parameters to capture these differential competitive effects will yield a negative alpha for Yodels versus Twinkies when the correct alpha should be positive or zero. Similarly, note that α_{23}^{1}, α_{23}^{4} and α_{23}^{5} from the third order solution are all negative. Although the effect is slight in each case, Yodels, Milky Way and Three Musketeers all pull more from Twinkies than Devil Dogs. Thus, this differential competitive effect causes the alpha for Twinkies versus Devil Dogs to be negative when the higher order alphas are not included.

Space limitations preclude a complete comparison of the 3rd order solution with the 4th order solution but notice that:

(1) all 18 of the 3rd order alphas from the 4th order model for which we have a priori

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TABLE 3

Comparison of Fit and Statistical Tests of Alternative Models: Group-Level Political Data

Order of Model	Number of Independent Parameters	Number of Independent Observations	Error Sum of Squares	R-Square	Adjusted R-Square	F with Previous Level	Probability for F
2	5	154	9.710	0.7993	0.7940	148.40	$p < 0.0001$
3	29	154	2.974	0.9385	0.9248	11.80	$p < 0.0001$
4	74	154	1.319	0.9727	0.9479	2.23	$p < 0.0009$
5	114	154	0.388	0.9920	0.9694	2.40	$p < 0.0033$
6	129	154	0.155	0.9968	0.9804	2.51	$p < 0.0202$

expectations are in the expected direction—the one exception from the 3rd order solution, α^3_{25}, went from 0.111 to −0.312; and

(2) in general, the 4th order alphas are closer to zero than the lower order alphas as would be expected given that the 4th order solution did not substantially improve the fit.

Group-Level Application

The Experiment and the Data. In this study the models were applied to data resulting from aggregated choices in a political context. The subjects were all volunteers, 40 graduate students who responded to recruiting pleas from the first author and a research assistant.[3] Each subject received a booklet which listed 66 choice sets. Fifty-seven of the sets were those formed by taking all possible subsets of size 2 or more from a basic set of 6 Presidential Candidates in the early primaries of the 1980 election—Bush, Brown, Carter, Connally, Kennedy, and Reagan. The remaining 9 of the 66 choice sets were randomly selected replicates, included to check on the stability of the vote shares. (The simple correlation between vote shares for the 9 repeated sets was 0.989.) For each choice set subjects were asked to "indicate the candidate for whom he/she would most likely vote if the primary were held today." The subjects' choices were aggregated to yield the proportion of votes obtained by each candidate in each of the 66 subsets.

Two characteristics of the data should be noted. First, it was collected in November 1979, before the primaries for the 1980 election. Second, the sample was composed of students in a graduate school of business, and therefore is not likely to yield results generalizable to any particular population of interest. The data set, however, is appropriate for demonstrating the empirical validity and practical usefulness of the models.

Expected Results

As in the individual-level application it was expected that differential inter-product competitive effects would cause the 3rd order (or higher) models to fit significantly better than the 2nd order model. In contrast to the individual-level application, the political candidates did not necessarily fall into 2 clearly defined submarkets. As a consequence, it was not expected that the improvement in fit from the 2nd to the 3rd order model would be of the same magnitude as in the individual-level application.

Actual Results

As shown in Table 3, the 3rd order model fits significantly better than the 2nd ($p < 0.0001$), although the increase in adjusted R^2 (0.7940 to 0.9248) is not of the magnitude observed in the case of the individual-level data. For this data set the 4th order

[3] The authors would like to thank Barry Friedman for his help in the collection, coding, and keypunching of the data reported here.

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TABLE 4

Parameter Estimates from the 3rd Order Model: Group-Level, Political Data

	Scaled Second Order Effects*														
Parameter	i/j = 1/2	1/3	1/4	1/5	1/6	2/3	2/4	2/5	2/6	3/4	3/5	3/6	4/5	4/6	5/6
$\alpha_{ij}^{\varnothing}$	0.470	−0.301	0.223	−0.198	0.588	−0.892	−0.139	−1.050	0.548	0.365	0.122	1.015	−0.274	0.599	0.737
	Scaled Third Order Effects														
Parameter	i/j = 1/2	1/3	1/4	1/5	1/6	2/3	2/4	2/5	2/6	3/4	3/5	3/6	4/5	4/6	5/6
α_{ij}^{k}															
k = 1	—	—	—	—	—	0.010	−0.174	−0.261	−0.139	−0.186	−0.261	−0.151	−0.083	0.039	0.113
k = 2	—	0.182	−0.117	−0.062	0.095	—	—	—	—	−0.301	−0.248	−0.083	0.049	0.207	0.166
k = 3	0.293	—	−0.248	0.030	−0.030	—	−0.545	−0.261	−0.329	—	—	—	0.278	0.215	−0.062
k = 4	−0.041	0.086	—	−0.236	0.405	0.131	—	−0.198	0.451	—	−0.329	0.315	—	—	0.652
k = 5	0.565	0.270	−0.301	—	−0.041	−0.288	−0.868	—	−0.616	−0.580	—	−0.315	—	0.262	—
k = 6	−0.186	−0.010	0.140	−0.105	—	0.182	0.329	0.086	—	0.157	−0.094	—	−0.248	—	—

*1 = Bush 3 = Carter 5 = Kennedy
2 = Brown 4 = Connally 6 = Reagan

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model fits significantly better than the 3rd ($p < 0.0009$) but note that the adjusted R^2 only increases from 0.9248 to 0.9479, a comparatively small increase when contrasted with the 0.7940 to 0.9248 in going from the 2nd order to the 3rd order model. Again, for reasons of parsimony and ease in exposition, suppose we adopt the 3rd order model which explains 94% of the variance in the market shares and has a 5.3 observation-to-parameter ratio.

Consider, now, Table 4 which contains parameter estimates for the 3rd order model. The second order alphas demonstrate that Carter, for example, is preferred to every one of the other candidates when each is considered separately in a one-on-one competition.

The third order alphas, however, provide the most interesting results. Carter pulls dramatically more from Brown than Connally ($\alpha^3_{24} = -0.545$). As one might expect, Connally pulls much more strongly from Reagan than he does from Kennedy ($\alpha^4_{56} = 0.652$). The values for the third order alphas clearly demonstrate the source of the more significant political threats for the respective candidates. Additional insights into the sources of intercandidate competitive effects can be obtained from examination of the estimated third order alphas. Furthermore, in combination with the second-order alphas, the third-order alphas would indicate to a particular candidate just whom he would benefit the greatest from if he could engineer a withdrawal.

Conclusions and Future Research

We believe the proposed class of market share models will enjoy the greatest acceptance if, in general, it continues to be observed that the lower order (more parsimonious) models can account for most of the variance in market shares.

Multiattribute Generalizations

Clearly the next most important step would be one or more multiattribute versions of the models. Although the proposed models can be used to describe the competitive structure within a market, the ultimate goal is the *prediction* of market shares for new products before their introduction. To do this, a model's parameters must be linked to the product attributes which govern choice. We have proposed these nonattribute based models for four reasons. First, as already noted, it has descriptive value in assessing competitive effects such as substitutability and dominance. Second, if one of the tree models cited earlier is thought to have potential, but the product attributes governing choice in the market are not well known, or the hierarchy of the tree is not clear a priori, the parameters resulting from a fit of the proposed models could yield insight into the underlying competitive structure that would suggest both relevant attributes and potential tree structures. Third, a comparison of the fits of the different order solutions helps diagnose the extent to which the observed market shares deviate from the Luce model. But fourth, and perhaps most important, there may exist several potential (but competing) multiattribute extensions of the proposed models. Just as the Luce model spawned both the MCI (Huff, 1962) and MNL (McFadden, 1973) multiattribute extensions and applications, it is hoped that the proposed generalization of the Luce model will result in one or more multiattribute extensions as well.[4]

Appendix A

(A-1) There exist unique numbers α^I_{ij} defined for $i, j \in T$, $I \subset T$ with $I \cap \{i, j\} = \varnothing$ such that for every subset $A \subset T$

$$\beta^A_{ij} = \sum_{I \subset A - \{i,j\}} \alpha^I_{ij}. \tag{A-2}$$

$$\text{In fact } \alpha^I_{ij} = (-1)^{\#(I)} \sum_{J \subset I} (-1)^{\#(J)} \beta^{ijJ}_{ij}. \tag{A-3}$$

PROOF. We will define α^I_{ij} inductively on the cardinality of I, $\#(I)$ = number of elements of I.

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First, if $A = \{i, j\}$, $I = \varnothing$, we have $\beta_{ij}^{A} = \beta_{ij}^{ij} = \alpha_{ij}^{\varnothing}$.

Now suppose α_{ij}^{I} have been defined for $\#(I) \leq n-2$, so that (A-2) is true for all A with $\#(A) \leq n$. In order that (A-2) be true for the set $A = ijkB$ with $\#(A) = n+1$ and $A = \{i, j, k\} \cup B$, then

$$\begin{aligned}\beta_{ij}^{A} &= \sum_{I \subset kB} \alpha_{ij}^{I} \\ &= \sum_{I \subset B} \alpha_{ij}^{I} + \alpha_{ij}^{kI} \\ &= \beta_{ij}^{ijB} + \Big(\sum_{I \not\subseteq B} \alpha_{ij}^{kI}\Big) + \alpha_{ij}^{kB}.\end{aligned}$$

By induction α_{ij}^{kI} has been defined for $I \neq B$. Hence we must have

$$\alpha_{ij}^{kB} = \beta_{ij}^{A} - \beta_{ij}^{ijB} - \Big(\sum_{I \not\subseteq B} \alpha_{ij}^{kI}\Big). \tag{A-4}$$

Equation (A-4) shows that α_{ij}^{I} is unique if it exists.

To show existence we define α_{ij}^{I} by equation (A-3) and verify (A-2).

$$\begin{aligned}\sum_{I \subset A} \alpha_{ij}^{I} &= \sum_{I \subset [A-\{i,j\}]} (-1)^{\#(I)} \sum_{J \subset I} (-1)^{\#(J)} \beta_{ij}^{ijJ} \\ &= \sum_{J \subset [A-\{i,j\}]} (-1)^{\#(J)} \beta_{ij}^{ijJ} \sum_{J \subset I \subset [A-\{i,j\}]} (-1)^{\#(I)}.\end{aligned}$$

Let $n = \#(A - (\{i, j\} \cup J))$. Then

$$\begin{aligned}\sum_{J \subset I \subset A-\{i,j\}} (-1)^{\#(I)} &= \sum_{k=0}^{n} (-1)^{\#(J)+k} \binom{n}{k} \\ &= \begin{cases} (-1)^{\#(J)} & \text{if} \quad n = 0, \\ 0 & \text{if} \quad n > 0. \end{cases}\end{aligned}$$

Thus, $\sum_{I \subset A} \alpha_{ij}^{I} = \beta_{ij}^{A}$.

Appendix B

In this appendix we prove the relationship between the parameters noted in Proposition 9.

Relationship Between Parameters

The factors α_{ij}^{I} are not independent. The relations between them are implied by (1), (2), and (3) and are specified in the next result.

RESTATEMENT OF PROPOSITION 9.

If $I \cap \{i, j, k\} = \varnothing$ *then*

$$\alpha_{ij}^{I} + \alpha_{ji}^{I} = 0. \tag{B-1}$$

$$\alpha_{ij}^{I} + \alpha_{jk}^{I} + \alpha_{ki}^{I} + \alpha_{ij}^{kI} + \alpha_{jk}^{iI} + \alpha_{ki}^{jI} = 0. \tag{B-2}$$

PROOF. (B-1) follows immediately from (2) and (6). For (B-2) we notice that by (6)

$$\begin{aligned}\alpha_{ij}^{kI} &= (-1)^{\#(I)+1} \sum_{J \subset I} (-1)^{\#(J)} (\beta_{ij}^{ijJ} - \beta_{ij}^{ijkJ}) \\ &= -\alpha_{ij}^{I} + (-1)^{\#(I)} \sum_{J \subset I} (-1)^{\#(J)} \beta_{ij}^{ijkJ} \qquad \text{or} \\ \alpha_{ij}^{kI} + \alpha_{ij}^{I} &= -1^{\#(I)} \sum_{J \subset I} (-1)^{\#(J)} \beta_{ij}^{ijkJ}.\end{aligned} \tag{B-3}$$

Now (B-2) follows immediately from (3) and (B-3).

Appendix C

(C-1) THEOREM. *There are unique constants* s_i^I *defined for each nonempty subset* $I \subset T$ *and each* $i \in T$ *such that*

$$\beta_{ij}^{A} = \sum_{I \subset A} (s_i^I - s_j^I), \tag{C-2}$$

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$$s_i^I = 0, \qquad i \notin I, \tag{C-3}$$

$$\sum_{i \in I} s_i^{iI} = u_I, \tag{C-4}$$

where u_I is any preassigned number.

The s_i^I for $i \in I$ can be calculated inductively by the formulae

$$s_i^T = u_{T-\{i\}}, \qquad i \in T, \tag{C-5}$$

$$s_i^{iI} = 1/(N - \#(I))[u_I + u_{iI} + \sum_{j \in iI} (\alpha_{ij}^I - s_i^{ijI})] \tag{C-6}$$

if $i \notin I$ and $\#(I) \leq N - 2$.

PROOF. Suppose first that (C-2), (C-3), and (C-4) are satisfied. From (C-2) it follows that

$$\alpha_{ij}^I = s_i^{iI} + s_i^{ijI} - s_j^{jI} - s_j^{ijI} \qquad \text{if} \qquad I \cap \{i, j\} = \varnothing. \tag{C-7}$$

Hence

$$s_i^{iI} + s_i^{ijI} = \alpha_{ij}^I + s_j^{jI} + s_i^{ijI}. \tag{C-8}$$

Let $b = N - \#(I)$. Then by taking the sum of (C-8) over $j \notin iI$ plus s_i^{iI}, we get

$$b \cdot (s_i^{iI}) + \sum_{j \notin iI} s_i^{ijI} = \sum_{j \notin iI} \alpha_{ij}^I + \sum_{j \notin I} s_j^{iI} + \sum_{j \notin iI} s_j^{jiI}.$$

Using (C-4), this becomes

$$b \cdot (s_i^{iI}) + \sum_{j \notin iI} s^{ijI} = u_I + u_{iI} + \sum_{j \notin iI} \alpha_{ij}^I,$$

which is (C-6). Since (C-5) is just the special case of (C-4) with $I = T - \{i\}$, we have shown that the s_i^I are uniquely determined by (C-2), (C-3), and (C-4), and satisfy (C-5) and (C-6).

Now suppose that the s_i^I, $i \in I$ are given by (C-5) and (C-6). We must show that (C-2) and (C-4) are satisfied. First, (C-4) is true if $\#(I) = N - 1$ by (C-5). Suppose (C-4) is true if $\#(I) \geq n$. Then if $\#(I) = n - 1$ we have by (C-6)

$$\sum_{i \in I} s_i^{iI} = 1/(N - \#(I))[u_I + \sum_{j \in I} u_{iI} + \sum_{i,j \in I} (\alpha_{ij}^I - s_i^{ijI})]$$

$$= u_I$$

since $\alpha_{ij}^I + \alpha_{ji}^I = 0$ and $\sum_{i \notin jI} s_i^{ijI} = u_{jI}$.

Instead of proving (C-2), we will prove (C-7). Again we proceed by backwards induction on $\#(I)$. First suppose $\#(I) = N - 2$ and $I \cup \{i, j\} = T$. Then

$$s_i^{ijI} = s_i^T = u_{T-\{i\}} = u_{jI}, \qquad s_i^{iI} = \tfrac{1}{2}[u_I + u_{iI} + (\alpha_{ij}^I - u_{jI})],$$

so

$$s_i^{iI} + s_i^{ijI} - s_j^{jI} - s_j^{ijI} = \tfrac{1}{2}[u_I + 2u_{iI} + \alpha_{ij}^I - u_I - 2u_{jI} - \alpha_{ji}^I] + u_{jI} - u_{iI}$$

$$= \alpha_{ij}^I \qquad \text{using (9).}$$

Suppose the result is true if $\#(I) \geq n$ and suppose $\#(I) = n - 1$. Then,

$$s_i^{iI} + s_i^{ijI} - s_j^{jI} - s_j^{ijI} = 1/(N - \#(I))[u_{iI} - u_{jI} + \sum_k [(\alpha_{ik}^I - \alpha_{jk}^I) + (s_j^{jkI} - s_i^{ikI})]] + s_i^{ijI} - s_j^{ijI}$$

$$= 1/(N - \#(I))[\sum_k (s_k^{ikI} - s_k^{jkI}) + (s_j^{jkI} - s_i^{ikI}) + (s_i^{ijI} - s_j^{ikI}) + (\alpha_{ik}^I - \alpha_{jk}^I)].$$

But

$$s_k^{ikI} + s_j^{jkI} + s_i^{ijI} - s_k^{jkI} - s_i^{ikI} - s_j^{ijI} = (s_k^{ikI} + s_k^{ijkI} - s_j^{ijI} - s_j^{ijkI})$$

$$+ (s_i^{jkI} + s_j^{ijkI} - s_i^{ikI} - s_i^{ijkI}) + (s_i^{ijI} + s_i^{ijkI} - s_k^{jkI} - s_k^{ijkI})$$

$$= \alpha_{kj}^{iI} + \alpha_{ji}^{kI} + \alpha_{ik}^{jI} \qquad \text{by induction.}$$

Since $\alpha_{ik}^I + \alpha_{kj}^I + \alpha_{ik}^{jI} + \alpha_{kj}^{iI} + \alpha_{ji}^{kI} = \alpha_{ij}^I$ for all k, we have

$$s_i^{iI} + s_i^{ijI} - s_j^{jI} - s_j^{ijI} = \alpha_{ij}^I$$

as claimed and the theorem is proved.

REMARK. The scale values s_i^{iI} are given in terms of the effects α_{ij}^I and the normalization constants u_1 by

$$s_i^{iI} = \frac{1}{N - \#(I)} u_I + \sum_{I \subset J \subset T - \{i\}} \sum_{j \in iI} (-1)^{\#(J-I)} \frac{\#(J - I)!(N - \#(J) - 2)!}{(N - \#(I))!} (u_{iJ} - u_{jJ})$$

$$+ \sum_{I \subset J \subset T - \{i\}} \sum_{j \in iI} (-1)^{\#(J-I)} \frac{\#(J - I)!(N - \#(J) - 1)!}{(N - \#(I))!} \alpha_{ij}^J.$$

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While this formula is revealing it is probably not as efficient as the inductive formula (C-5) and (C-6) if one actually needs to calculate the scale values. The important special case when the normalization constants u_I are all zero is, of course, much simpler. It shows that for the nth order model (i.e. $\alpha_{ij}^I = 0$ if $\#(J) \geq n-1$) the scale values $s_i^I = 0$ if $\#(I) \geq n$.

It finally remains to show that

$$P_A(i) = \frac{\prod_{I \subset A} e^{s_i^I}}{\sum_{k \in A} \prod_{I \subset A} e^{s_k^I}}. \tag{C-9}$$

Since $\beta_{ij}^A = \log (P_A(i)/P_A(j))$, (C-2) can be written as

$$P_A(i) = P_A(j) \prod_{I \subset A} \frac{e^{s_i^I}}{e^{s_j^I}};$$

we now take the sum over $i \in A$ and get

$$1 = P_A(j) \sum_{i \in A} \prod_{I \subset A} \frac{e^{s_i^I}}{e^{s_j^I}}.$$

Thus,

$$P_A(i) = \frac{\prod_{I \subset A} \frac{e^{s_i^I}}{e^{s_j^I}}}{\sum_{k \in A} \prod_{I \subset A} \frac{e^{s_k^I}}{e^{s_j^I}}} = \frac{\prod_{I \subset A} e^{s_i^I}}{\sum_{k \in A} \prod_{I \subset A} e^{s_k^I}}.$$

Appendix D

$$P_A(i) = \frac{\prod_{I \subset A} e^{s_i^I}}{\sum_{k \in A} \prod_{I \subset A} e^{s_k^I}}, \tag{D-1}$$

$$\ln P_A(i) = \sum_{I \subset A} s_i^I - \ln \left[\sum_{k \in A} \prod_{I \subset A} e^{s_k^I} \right]. \tag{D-2}$$

Sum over all j in A and divide by N_A = # of elements in A.

$$\frac{1}{N_A} \sum_{j \in A} \ln P_A(j) = \frac{1}{N_A} \sum_{j \in A} \sum_{I \subset A} s_j^I - \frac{N_A}{N_A} \ln \left[\sum_{k \in A} \prod_{I \subset A} e^{s_k^I} \right].$$

Thus,

$$\ln \left[\sum_{k \in A} \prod_{I \subset A} e^{s_k^I} \right] = \frac{1}{N_A} \sum_{j \in A} \sum_{I \subset A} s_j^I - \frac{1}{N_A} \sum_{j \in A} \ln P_A(j).$$

So, substitute for $\ln [\sum_{k \in A} \prod_{I \subset A} e^{s_k^I}]$ in (D-2) which gives

$$\ln P_A(i) = \sum_{I \subset A} s_i^I - \frac{1}{N_A} \sum_{j \in A} \sum_{I \subset A} s_j^I + \frac{1}{N_A} \sum_{j \in A} \ln P_A(j).$$

Let $\tilde{P}_A = [\prod_{j \in A} P_A(j)]^{1/N_A}$ and note that

$$\ln \tilde{P}_A = \frac{1}{N_A} \sum_{j \in A} \ln P_A(j).$$

Thus,

$$\ln \left[\frac{P_A(i)}{\tilde{P}_A} \right] = \frac{N_A - 1}{N_A} \sum_{I \subset A} s_i^I - \frac{1}{N_A} \sum_{\substack{j \in A \\ j \neq i}} \sum_{I \subset A} s_j^I. \tag{D-3}$$

(D-3) is linear in the parameters, s_i^I, and least-squares can be used to estimate the parameters.

The s_i^I's are not unique and a series of normalizing equations must be included as "pseudo-observations" to guarantee uniqueness. These equations can be of the form $0 = \sum_{i \in I} s_i^{iI}$.

Appendix E. Example of Parameter Estimation on a Hypothetical Set of Data

This appendix demonstrates parameter estimation using a hypothetical set of data. Because the data have been constructed to illustrate substitutability and dominance problems, the appendix shows how the models handle these problems as well. (In Appendix F we show how the models accommodate violations of regularity.)

Assume Table E-1 displays the market shares for all possible subsets of 4 products. These market shares might be found, for example, if the 4 products were Pepsi (1), Coke (2), 7-Up (3) and Sprite (4), respectively and consumers whose choices generated the data:

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TABLE E-1

Market Shares for All Possible Subsets of 4 Products: Hypothetical Data

Products in Choice Set	Market Shares for Products in Set			
	Product 1	Product 2	Product 3	Product 4
1. (1, 2)	0.98	0.02		
2. (1, 3)	0.50		0.50	
3. (1, 4)	0.50			0.50
4. (2, 3)		0.50	0.50	
5. (2, 4)		0.50		0.50
6. (3, 4)			0.90	0.10
7. (1, 2, 3)	0.49	0.01	0.50	
8. (1, 2, 4)	0.49	0.01		0.50
9. (1, 3, 4)	0.50		0.45	0.05
10. (2, 3, 4)		0.50	0.45	0.05
11. (1, 2, 3, 4)	0.49	0.01	0.45	0.05

TABLE E-2

	0	1	2	3	4	5	6	7	8	9	10	11	12	13
	$\ln\left(\frac{P_A(i)}{P_A}\right)$	s_1^1	s_2^2	s_3^3	s_4^4	s_1^{12}	s_2^{12}	s_1^{13}	s_3^{13}	s_1^{14}	s_4^{14}	s_2^{23}	s_3^{23}	s_2^{24}
1.	1.946 =	0.500	−0.500	0.	0.	0.500	−0.500	0.	0.	0.	0.	0.	0.	0.
2.	−1.946 =	−0.500	0.500	0.	0.	−0.500	0.500	0.	0.	0.	0.	0.	0.	0.
3.	0.000 =	0.500	0.	−0.500	0.	0.	0.	0.500	−0.500	0.	0.	0.	0.	0.
4.	0.000 =	−0.500	0.	0.500	0.	0.	0.	−0.500	0.500	0.	0.	0.	0.	0.
5.	0.000 =	0.500	0.	0.	−0.500	0.	0.	0.	0.	0.500	−0.500	0.	0.	0.
6.	0.000 =	−0.500	0.	0.	0.500	0.	0.	0.	0.	−0.500	0.500	0.	0.	0.
7.	0.000 =	0.	0.500	−0.500	0.	0.	0.	0.	0.	0.	0.	0.500	−0.500	0.
8.	0.000 =	0.	−0.500	0.500	0.	0.	0.	0.	0.	0.	0.	−0.500	0.500	0.
9.	0.000 =	0.	0.500	0.	−0.500	0.	0.	0.	0.	0.	0.	0.	0.	0.500
10.	0.000 =	0.	−0.500	0.	0.500	0.	0.	0.	0.	0.	0.	0.	0.	−0.500
11.	1.099 =	0.	0.	0.500	−0.500	0.	0.	0.	0.	0	0.	0.	0.	0.
12.	−1.099 =	0.	0.	−0.500	0.500	0.	0.	0.	0.	0.	0.	0.	0.	0.
13.	1.291 =	0.667	−0.333	−0.333	0.	0.667	−0.333	0.667	−0.333	0.	0.	−0.333	−0.333	0.
14.	−2.601 =	−0.333	0.667	−0.333	0.	−0.333	0.667	−0.333	−0.333	0.	0.	0.667	−0.333	0.
15.	1.311 =	−0.333	−0.333	0.667	0.	−0.333	−0.333	−0.333	0.667	0.	0.	−0.333	0.667	0.
16.	1.291 =	0.667	−0.333	0.	−0.333	0.667	−0.333	0.	0.	0.667	−0.333	0.	0.	−0.333
17.	−2.600 =	−0.333	0.667	0.	−0.333	−0.333	0.667	0.	0.	−0.333	−0.333	0.	0.	0.667
18.	1.311 =	−0.333	−0.333	0.	0.667	−0.333	−0.333	0.	0.	−0.333	0.667	0.	0.	−0.333
19.	0.803 =	0.667	0.	−0.333	−0.333	0.	0.	0.667	−0.333	0.667	−0.333	0.	0.	0.
20.	0.697 =	−0.333	0.	0.667	−0.333	0.	0.	−0.333	0.667	−0.333	−0.333	0.	0.	0.
21.	−1.500 =	−0.333	0.	−0.333	0.667	0.	0.	−0.333	−0.333	−0.333	0.667	0.	0.	0.
22.	0.803 =	0.	0.667	−0.333	−0.333	0.	0.	0.	0.	0.	0.	0.667	−0.333	0.667
23.	0.697 =	0.	−0.333	0.667	−0.333	0.	0.	0.	0.	0.	0.	−0.333	0.667	−0.333
24.	−1.500 =	0.	−0.333	−0.333	0.667	0.	0.	0.	0.	0.	0.	−0.333	−0.333	−0.333
25.	1.565 =	0.750	−0.250	−0.250	−0.250	0.750	−0.250	0.750	−0.250	0.750	−0.250	−0.250	−0.250	−0.250
26.	−2.327 =	−0.250	0.750	−0.250	−0.250	−0.250	0.750	−0.250	−0.250	−0.250	−0.250	0.750	−0.250	0.750
27.	1.480 =	−0.250	−0.250	0.750	−0.250	−0.250	−0.250	−0.250	0.750	−0.250	−0.250	−0.250	0.750	−0.250
28.	−0.718 =	−0.250	−0.250	−0.250	0.750	−0.250	−0.250	−0.250	−0.250	−0.250	0.750	−0.250	−0.250	−0.250
29.	0.000 =	1.0	1.0	1.0	1.0	0.	0.	0.	0.	0.	0.	0.	0.	0.
30.	0.000 =	0.	0.0	0.	0.	0.	1.0	0.	1.0	0.	1.0	0.	0.	0.
31.	0.000 =	0.	0.	0.	0.	1.0	0.	0.	0.	0.	0.	0.	1.0	0.
32.	0.000 =	0.	0.	0.	0.	0.	0.	1.0	0.	0.	0.	1.0	0.	0.
33.	0.000 =	0.	0.	0.	0.	0.	0.	0.	0.	1.0	0.	0.	0.	1.0
34.	0.000 =	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
35.	0.000 =	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
36.	0.000 =	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
37.	0.000 =	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
38.	0.000 =	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
39.	0.000 =	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.

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(1) viewed the products as essentially a choice between a cola and a noncola;
(2) almost always chose Pepsi over Coke when both are available and they want a Cola; and
(3) almost always chose 7-Up over Sprite when both are available and they want a noncola.

Least-squares can be used to estimate the parameters, s_i^I, given by (12) using the transformation described in Appendix D. Table E-2 contains the simultaneous equations given by (D-3). Column 0 of Table E-2 contains the value for the dependent variable in a vector we reference here as Y. Columns 1 through 28 contain the values for the independent variables s_i^I in a matrix we reference here as X. Rows 1 through 28 correspond to the observed market shares. Rows 29 through 39 contain the normalizing equations necessary to guarantee uniqueness of the s_i^I's. Specifically, row 29 contains the normalizing equations for the s_i^I's where $\#(I) = 1$, rows 30 through 33 contain the equations for the s_i^I's where $\#(I) = 2$, and rows 34 through 39 contain the equations for the s_i^I's where $\#(I) = 3$.

Let S be a vector containing the s_i^I's. The fourth-order solution is given by

$$S = (X'X)^{-1}X'Y \text{ where } Y \text{ contains all 39 rows and } X \text{ is all 28 columns by all 39 rows.} \quad \text{(E-1)}$$

The third-order solution is given by (E-1) where Y contains rows 1–33 and X contains columns 1–16 and rows 1–33.

The second-order solution is given by (E-1) where Y contains rows 1–29 and X contains columns 1–4 and rows 1–29.

Table E-3 describes the relative fit of each of the 3 solutions. Table E-4 contains the s_i^I's which result from each of the 3 solutions. Table E-5 contains the α_{ij}^I's given by the respective s_i^I's using equation (16).

14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
s_4^{24}	s_3^{34}	s_4^{34}	s_1^{123}	s_2^{123}	s_3^{123}	s_1^{124}	s_2^{124}	s_4^{124}	s_1^{134}	s_3^{134}	s_4^{134}	s_2^{234}	s_3^{234}	s_4^{234}
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
−0.500	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.500	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.500	−0.500	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	−0.500	0.500	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.667	−0.333	−0.333	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	−0.333	0.667	−0.333	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	−0.333	−0.333	0.667	0.	0.	0.	0.	0.	0.	0.	0.	0.
−0.333	0.	0.	0.	0.	0.	0.667	−0.333	−0.333	0.	0.	0.	0.	0.	0.
−0.333	0.	0.	0.	0.	0.	−0.333	0.667	−0.333	0.	0.	0.	0.	0.	0.
0.667	0.	0.	0.	0.	0.	−0.333	−0.333	0.667	0.	0.	0.	0.	0.	0.
0.	−0.333	−0.333	0.	0.	0.	0.	0.	0.	0.667	−0.333	−0.333	0.	0.	0.
0.	0.667	−0.333	0.	0.	0.	0.	0.	0.	−0.333	0.667	−0.333	0.	0.	0.
0.	−0.333	0.667	0.	0.	0.	0.	0.	0.	−0.333	−0.333	0.667	0.	0.	0.
−0.333	−0.333	−0.333	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.667	−0.333	−0.333
−0.333	0.667	−0.333	0.	0.	0.	0.	0.	0.	0.	0.	0.	−0.333	0.667	−0.333
0.667	−0.333	0.667	0.	0.	0.	0.	0.	0.	0.	0.	0.	−0.333	−0.333	0.667
−0.250	−0.250	−0.250	0.750	−0.250	−0.250	0.750	−0.250	−0.250	0.750	−0.250	−0.250	−0.250	−0.250	−0.250
−0.250	−0.250	−0.250	−0.250	0.750	−0.250	−0.250	0.750	−0.250	−0.250	−0.250	−0.250	0.750	−0.250	−0.250
−0.250	0.750	−0.250	−0.250	−0.250	0.750	−0.250	−0.250	−0.250	−0.250	0.750	−0.250	−0.250	0.750	−0.250
0.750	−0.250	0.750	−0.250	−0.250	−0.250	−0.250	−0.250	0.750	−0.250	−0.250	0.750	−0.250	−0.250	0.750
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
1.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	1.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	1.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	1.0	0.	0.	1.0	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	1.0	0.	0.	0.	0.	0.	0.	1.0	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	1.0	0.	0.	1.0	0.	0.	0.	0.
0.	0.	0.	1.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	1.0
0.	0.	0.	0.	0.	0.	1.0	0.	0.	0.	0.	0.	0.	1.0	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.	1.0	0.	0.	1.0	0.	0.

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TABLE E-3

Comparison of Fit of Alternative Models: Hypothetical Data*

Order of Model	Number of Independent Parameters	Number of Independent Observations	Error Sum of Squares	R-Square	Adjusted R-Square
2	3	17	18.86	0.604	0.547
3	11	17	0.00	1.000	1.000
4	17	17	0.00	1.000	1.000

* No F-tests comparing models of order 4 to 3 and 3 to 2 can be done because the error sum-of-squares for the 4th and 3rd order solutions are 0.

TABLE E-4

Fourth-Order Solution—Values of S_i^I For:

i Equal To:	And I Equal To: {1}	{2}	{3}	{4}	{1, 2}	{1, 3}	{1, 4}	{2, 3}	{2, 4}	{3, 4}	{1, 2, 3}	{1, 2, 4}	{1, 3, 4}	{2, 3, 4}
1	0.776	—	—	—	−0.013	0.768	0.035	—	—	—	0.	0.	0.	—
2	—	−0.522	—	—	−2.608	—	—	0.768	0.035	—	0.	0.	—	0.
3	—	—	0.239	—	—	1.304	—	0.007	—	−0.070	0.	—	0.	0.
4	—	—	—	−0.493	—	—	1.304	—	0.007	−1.535	—	0.	0.	0.

Third-Order Solution—Values of S_i^I For:

i Equal To:	And I Equal To: {1}	{2}	{3}	{4}	{1, 2}	{1, 3}	{1, 4}	{2, 3}	{2, 4}	{3, 4}
1	0.776	—	—	—	−0.013	0.768	0.035	—	—	—
2	—	−0.522	—	—	−2.608	—	—	0.768	0.035	—
3	—	—	0.239	—	—	1.304	—	0.007	—	−0.070
4	—	—	—	−0.493	—	—	1.304	—	0.007	−1.535

Second-Order Solution—Values of S_i^I For:

i Equal To:	And I Equal To: {1}	{2}	{3}	{4}
1	1.217	—	—	—
2	—	−1.531	—	—
3	—	—	0.932	—
4	—	—	—	−0.619

TABLE E-5

Fourth-Order Solution—Values of α_{ij}^I For:

i/j Equal To:	And I Equal To: {∅}	{1}	{2}	{3}	{4}	{1, 2}	{1, 3}	{1, 4}	{2, 3}	{2, 4}	{3, 4}
1/2	3.892	—	—	0.000	0.000	—	—	—	—	—	0.000
1/3	0.000	—	−0.020	—	0.105	—	—	—	—	0.000	—
1/4	0.000	—	−0.020	2.303	—	—	—	—	0.000	—	—
2/3	0.000	−3.912	—	—	0.150	—	—	0.000	—	—	—
2/4	0.000	−3.912	—	2.303	—	—	0.000	—	—	—	—
3/4	2.197	0.000	0.000	—	—	0.000	—	—	—	—	—

Third-Order Solution—Values of α_{ij}^I For:

i/j Equal To:	And I Equal To: {∅}	{1}	{2}	{3}	{4}
1/2	3.892	—	—	0.000	0.000
1/3	0.000	—	−0.020	—	0.105
1/4	0.000	—	−0.020	2.303	—
2/3	0.000	−3.912	—	—	0.105
2/4	0.000	−3.912	—	2.303	—
3/4	2.197	0.000	0.000	—	—

Second-Order Solution—Values of α_{ij}^I For:

i/j Equal To:	And I Equal To: {∅}
1/2	2.747
1/3	0.284
1/4	1.835
2/3	−2.463
2/4	−0.912
3/4	1.551

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TABLE F-1

Products in Choice Set	Market Shares for Products in Set			
	Product 1	Product 2	Product 3	Product 4
1. (1, 2)	0.5	0.5		
2. (1, 3)	0.5		0.5	
3. (1, 4)	0.5			0.5
4. (2, 3)		0.5	0.5	
5. (2, 4)		0.5		0.5
6. (3, 4)			0.5	0.5
7. (1, 2, 3)	1/3	1/3	1/3	
8. (1, 2, 4)	1/3	1/3		1/3
9. (1, 3, 4)	3/5		1/5	1/5
10. (2, 3, 4)		3/5	1/5	1/5
11. (1, 2, 3, 4)	3/8	3/8	1/8	1/8

Appendix F

We will now present a hypothetical case which does not have the regularity property to show how the third order model can handle lack of regularity.

Assume we have four products (labelled 1, 2, 3, and 4) all of equal preference, except that products 3 and 4 are very similar in a context where the presence of two similar but different products reduces the desirability of each of them. The market shares for all possible subsets could be as indicated in Table F-1. We have $\alpha_{ij}^{I} = 0$ except for $\alpha_{13}^{4} = \alpha_{14}^{3} = \alpha_{23}^{4} = \alpha_{24}^{3} = \ln 3$. Thus the third order effects completely account for the choice probabilities. Scale values:

$$S_1^1 = S_2^2 = -S_3^3 = -S_4^4 = \tfrac{1}{6} \ln 3, \qquad S_1^{12} = S_2^{12} = S_3^{13} = S_4^{14} = S_3^{23} = S_4^{24} = 0,$$

$$S_1^{13} = S_1^{14} = S_2^{23} = S_2^{24} = \tfrac{1}{3} \ln 3, \qquad S_3^{34} = S_4^{34} = -\tfrac{2}{3} \ln 3.$$

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